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(i) The average kinetic energy of the molecule is directly proportional to absolute temperature of the gas. The equation (9.9) gives the connection between the macroscopic world (temperature) to microscopic world (motion of molecules).

(ii) The average kinetic energy of each molecule depends only on temperature of the gas not on mass of the molecule. In other words, if the temperature of an ideal gas is measured using thermometer, the average kinetic energy of each molecule can be calculated without seeing the molecule through naked eye.

By multiplying the total number of gas molecules with average kinetic energy of each molecule, the internal energy of the gas is obtained.

Internal energy of ideal gas $U = N\left(\frac{1}{2}m\overline{v^2}\right)$

By using equation (9.9)

$$U = \frac{3}{2}NkT \qquad (9.11)$$

From equation (9.11), we understand that the internal energy of an ideal gas depends only on absolute temperature and is independent of pressure and volume.

EXAMPLE 9.1

A football at 27°C has 0.5 mole of air molecules. Calculate the internal energy of air in the ball.

Solution

The internal energy of ideal gas = $\frac{3}{2}NkT$.

The number of air molecules is given in terms of number of moles so, rewrite the expression as follows

$$U = \frac{3}{2}\mu RT$$

Since $Nk = \mu R$. Here μ is number of moles.

Gas constant $R = 8.31\ \frac{J}{mol\,k}$

Temperature T =273+27=300K

$U = \frac{3}{2} \times 0.5 \times 8.31 \times 300 = 1869.75J$

This is approximately equivalent to the kinetic energy of a man of 57 kg running with a speed of 8 m s^{-1}.

9.2.3 Relation between pressure and mean kinetic energy

From earlier section, the internal energy of the gas is given by

$$U = \frac{3}{2}NkT$$

The above equation can also be written as

$$U = \frac{3}{2}PV$$

since $PV = NkT$

$$P = \frac{2}{3}\frac{U}{V} = \frac{2}{3}u \qquad (9.12)$$

From the equation (9.12), we can state that the pressure of the gas is equal to two thirds of internal energy per unit volume or internal energy density ($u = \frac{U}{V}$).

Writing pressure in terms of mean kinetic energy density using equation (9.6)

$$P = \frac{1}{3}nm\overline{v^2} = \frac{1}{3}\rho\overline{v^2} \qquad (9.13)$$

where $\rho = nm$ = *mass density* (Note n is number density)

Multiply and divide R.H.S of equation (9.13) by 2, we get

$$P = \frac{2}{3}\left(\frac{\rho}{2}\overline{v^2}\right)$$

$$P = \frac{2}{3}\overline{KE} \qquad (9.14)$$